RESEARCH METHODOLOGY IN COMPUTER SCIENCE

WORKSHEET – UNIT III

Literature Search, Analysis, References & Ethics

Literature Search and Academic Sources

Databases → Search Strategies → Literature Analysis → Synthesis → Mapping

Academic Integrity and Research Ethics

References → Citation → Plagiarism → Data Privacy → Ethical Research

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UNIT III – WORKSHEET MAP

Literature Search, Analysis & Ethics

LEARN & REMEMBER

This unit develops the practical skills required to move from a research question to a reliable body of literature and then use that literature ethically.
The major learning areas are:

- Literature search and academic sources
- Research databases and search strategies
- Keywords and Boolean operators
- Citation chaining and search alerts
- Critical reading of research papers
- Literature analysis and synthesis
- Literature mapping and research trends
- References and citation management
- Plagiarism and effective paraphrasing
- Research ethics, data privacy and authorship
- IRB and ethical approval

No.	Learning Area	Done	Confidence
1	Literature Search	☑	5
2	Sources of Literature	☑	4
3	Literature Review	☑	5
4	Databases	☑	4
5	Search Strategies	☑	4
6	Citation Chaining	☑	5
7	Literature Analysis	☑	4
8	Literature Synthesis	☑	4
9	Literature Mapping	☑	4
10	References and Citations	☑	5
11	Plagiarism and Paraphrasing	☑	4
12	Research Ethics	☑	4
13	Data Privacy and IRB	☑	5
14	Authorship and Publication Ethics	☑	4

WARM-UP

Start Thinking Like a Researcher

PRACTICE ZONE

Think about your final-year project, research interest, or a topic you would like to investigate.

1. Where would you search for related work?
2. Would you use Google, Wikipedia, IEEE Xplore, ACM DL or something else?
3. How would you decide whether a paper is reliable?

My research topic: ML-based network intrusion detection (reducing false positives).

Where I would search first: IEEE Xplore.

Why?

It indexes peer-reviewed CS and security research with strong search filters.

1. LITERATURE SEARCH

Building the Research Foundation

1.1 What is Literature Search?

Literature search is the systematic process of finding existing research related to a research problem, question, method, technology, or domain.

It is much more than simply typing a topic into a search engine.

A good literature search helps the researcher discover:

- what is already known,
- what methods have already been used,
- which datasets and benchmarks are available,
- what results have already been achieved,
- where researchers disagree, and
- what questions remain unanswered.

LEARN & REMEMBER

Research does not start with "What can I build?"
It should first ask:

What has already been done, what is missing, and what can I contribute?

1.2 Why Literature Search Matters

Literature search:

1. helps avoid unnecessary duplication;
2. supports the novelty of a research problem;
3. identifies useful tools, datasets and benchmarks;
4. provides background for the research problem;
5. helps formulate research questions;
6. provides evidence for the proposed methodology;
7. identifies gaps and unresolved issues.

THINK LIKE A RESEARCHER

Suppose a student proposes:

"I will develop a CNN model for plant disease detection."

What should the student search before starting the work?

Existing plant-disease datasets, prior CNN methods, reported accuracies, and remaining gaps.

2. SOURCES OF LITERATURE

Academic vs. Casual Information

2.1 Primary and Secondary Sources

Source Type	Examples
Primary Sources	Original journal papers, conference papers, patents, technical reports
Secondary Sources	Review papers, survey papers, books
Digital Libraries	IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect
Indexing Databases	Scopus, Web of Science
Preprint Sources	arXiv and other research repositories
Search Engines	Google Scholar and general web search

2.2 Academic Source vs. Casual Source

Consider the following:

Casual Search	Academic Search
Blogs	Journal papers
News articles	Conference papers
YouTube videos	Technical reports
Wikipedia	Patents
Random websites	Peer-reviewed literature

COMMON TRAP

Blogs, videos and Wikipedia may be useful for learning a concept, but they should not automatically be treated as equivalent to peer-reviewed research evidence.

2.3 Mini Case Study

A student wants to research:

Blockchain for Supply Chain

The student first searches the general web and finds blogs and news articles.

After searching IEEE Xplore and other academic databases, the student finds research discussing scalability, energy consumption and security.

THINK LIKE A RESEARCHER

What changed between the two searches?

The academic search returned peer-reviewed evidence (scalability, energy, security) instead of blogs.

3. LITERATURE REVIEW

From Papers to Understanding

3.1 Objectives of a Literature Review

A literature review should not become a list of paper summaries.
It should:

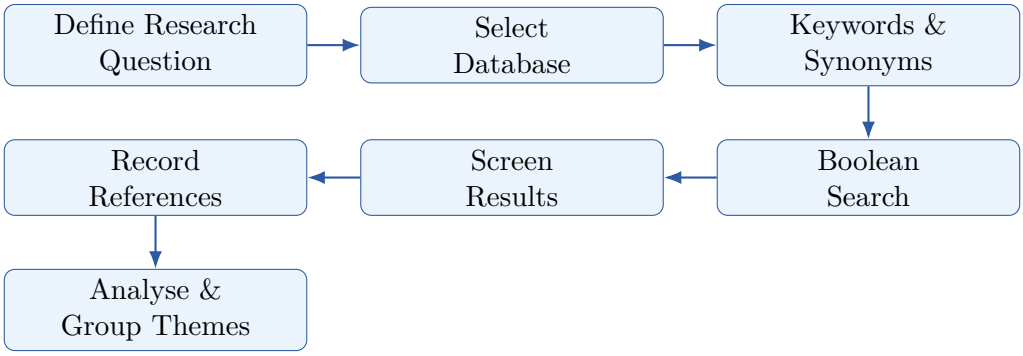
- summarize current knowledge,
- compare methods and results,
- identify research gaps,
- highlight contradictions or debates,
- establish the theoretical and methodological foundation of the study.

LEARN & REMEMBER

Weak literature review:
"Paper A did this. Paper B did this. Paper C did this."

Stronger literature review:
"Most studies use machine learning for intrusion detection; however, real-time deployment remains insufficiently explored because of computational and scalability constraints."

3.2 Literature Search Process



PRACTICE ZONE

Write the correct order for your own literature search.
Research Question: Can feature selection reduce false positives in ML-based NIDS?
Database: IEEE Xplore.
Keywords: intrusion detection, machine learning, false positives.
Search String: ("intrusion detection") AND ("machine learning") AND ("false positive")
How will I screen the papers?

By checking titles and abstracts for relevance, recency, and venue quality.

4. RESEARCH DATABASES

Finding Reliable Literature

4.1 Why Use Academic Databases?

Academic databases provide structured access to scholarly research.
They commonly support:

- author searching,
- year filtering,
- keyword searching,
- subject filtering,
- citation tracking,
- full-text access,
- export of citation information.

4.2 Major Databases in Computer Science

Database	Typical Coverage
IEEE Xplore	Computer science, electronics, AI, IoT, robotics, communications
ACM Digital Library	Computing, algorithms, software, HCI
SpringerLink	Computer science, engineering and interdisciplinary research
ScienceDirect	Broad scientific and engineering literature
Scopus	Abstracting, indexing and citation analysis
Web of Science	Indexing and citation analysis
arXiv	Preprints in AI, ML, mathematics and related areas
Google Scholar	Broad academic search with less filtering

PRACTICE ZONE

For each topic, select a suitable database.

1. Robotics and automation: IEEE Xplore.
2. Algorithms and software: ACM Digital Library.
3. AI and machine learning: arXiv / IEEE Xplore.
4. Citation analysis: Scopus / Web of Science.

5. KEYWORD SELECTION

Search with Precision

5.1 What Makes a Good Keyword?

A good keyword should be:

- technically relevant,
- specific to the research problem,
- commonly used in the field,
- supported by synonyms or alternative terminology.

5.2 Example

Research topic:

AI in Cybersecurity

Possible keywords:

Artificial Intelligence

Machine Learning

Intrusion Detection

Malware Detection

Deep Learning

LEARN & REMEMBER

A useful technique is to inspect the keywords used in good papers that you have already found. Those keywords can then be used to expand your search.

PRACTICE ZONE

Develop keywords for:

Topic: Drone Applications in Agriculture

Keyword 1	Drone / UAV
Keyword 2	Precision agriculture
Keyword 3	Crop monitoring
Synonym	Unmanned aerial vehicle
Related term	Remote sensing

5.3 Search Vocabulary

Think beyond one word.
For example:

IoT Security

may be expanded into:

IoT | Internet of Things | IoT Security | Network Security | Cybersecurity

6. BOOLEAN SEARCH

AND, OR, NOT and Advanced Search

6.1 Boolean Operators

Operator	Purpose	Example
AND	Narrows the search	AI AND Healthcare
OR	Broadens the search	IoT OR "Internet of Things"
NOT	Excludes a term	Cloud NOT Gaming
" "	Searches an exact phrase	"deep reinforcement learning"
*	Wildcard / word variation	comput*

6.2 Understanding the Difference

Suppose the researcher searches:

$$AI$$

The result may be very broad.
Now:

$$AI\ AND\ Healthcare$$

becomes more focused.
Now:

(Artificial Intelligence OR Machine Learning)AND(Healthcare OR Medical Diagnosis)

captures multiple related terms.

PRACTICE ZONE

Create a Boolean search string for:

Machine Learning for Crop Disease Detection

Your search string:

("machine learning" OR "deep learning") AND ("crop disease" OR "plant disease detection")

7. SEARCH STRATEGY

Search, Screen, Record

7.1 Screening Search Results

Do not download every result.
First examine:

1. Title
2. Abstract
3. Keywords
4. Publication year
5. Venue
6. Relevance to your research question

Only then decide whether the full paper deserves detailed reading.

COMMON TRAP

A common mistake is downloading hundreds of papers and reading none of them carefully. Quality of analysis is more important than the number of PDFs collected.

7.2 Search Record

Maintain a simple search record.

Date	Database	Search Query	Results / Notes

7.3 Common Search Mistakes

- Using only one or two keywords.
- Relying only on general web search.
- Ignoring Boolean operators.
- Not recording search strategies.
- Skipping abstracts.
- Collecting too many irrelevant papers.

8. CITATION CHAINING

Finding Connected Research

8.1 Backward Citation Chaining

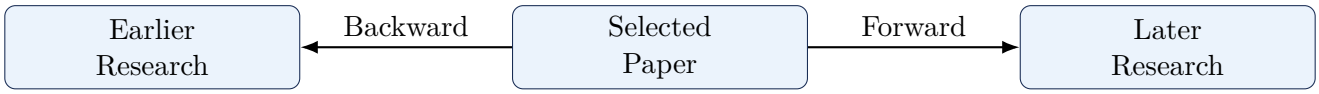
Backward chaining means examining the references of an existing paper.
It answers:

Which earlier studies influenced this paper?

8.2 Forward Citation Chaining

Forward chaining means finding newer papers that cite the selected paper.
It answers:

What research came after this work?



PRACTICE ZONE

You find one highly relevant paper.

1. Where will you look for older foundational studies?
2. Where will you look for newer developments?

Answer:

Backward chaining (references) for older studies; forward chaining (citing papers) for newer ones.

8.3 Search Alerts

Many databases allow researchers to create alerts for:

- new papers containing selected keywords,
- new publications by an author,
- new citations of a paper,
- new research in a particular topic.

THINK LIKE A RESEARCHER

If your research project lasts six months, how could search alerts help you?

They notify me of new relevant papers and citations, keeping the review current throughout the project.

9. CRITICAL READING OF A RESEARCH PAPER

Read Beyond the Abstract

9.1 Why Critical Reading?

Collecting papers is not enough.
Critical reading helps determine:

- whether the study actually addresses your problem,
- whether the methodology is appropriate,
- whether the dataset is suitable,
- whether the results support the claims,
- what limitations remain.

9.2 Paper Reading Sequence

Section	Questions to Ask
Abstract	What problem and contribution are presented?
Introduction	Why is the problem important? What research question is addressed?
Methodology	What algorithm, dataset or experimental design is used?
Results	What evidence is presented? What are the performance measures?
Discussion	What do the results actually mean?
Limitations	What did the authors not solve?
References	Which earlier studies are important?

9.3 Paper Analysis Sheet

Research Problem	High false-positive rates in network intrusion detection.
Research Question	Can ML cut false positives while keeping recall high?
Method / Algorithm	Random Forest classifier on flow features.
Dataset / Experimental Setup	CICIDS2017; 70/30 train-test split.
Main Result	High detection with a low false-positive rate.
Performance Metric	Accuracy, precision, recall, FP-rate.
Limitation	Evaluated on one dataset; offline only.
Future Scope	Real-time, on-device deployment and new attacks.

10. LITERATURE ANALYSIS

Evaluate Individual Studies

10.1 Analysis vs. Summary

A summary tells what a paper contains.
Analysis examines:

- strengths,
- weaknesses,
- methods,
- datasets,
- results,
- assumptions,
- limitations.

LEARN & REMEMBER

Summary:
"Paper A uses SVM for intrusion detection."

Analysis:
"Paper A uses SVM with the KDD Cup 99 dataset and achieves 92% accuracy. The method is computationally simple, but the dataset may not represent modern network traffic."

10.2 Comparative Analysis

Consider three studies:

	Paper A	Paper B	Paper C
Method	SVM	Deep Learning	Hybrid
Dataset	KDD Cup 99	NSL-KDD	NSL-KDD
Accuracy	92%	96%	High
Complexity	Low	Medium	High
Scalability	Good	Moderate	Limited

THINK LIKE A RESEARCHER

Which method would you investigate further if:

1. accuracy is the main concern?
2. computational cost is the main concern?
3. real-time deployment is the main concern?

Explain your reasoning.

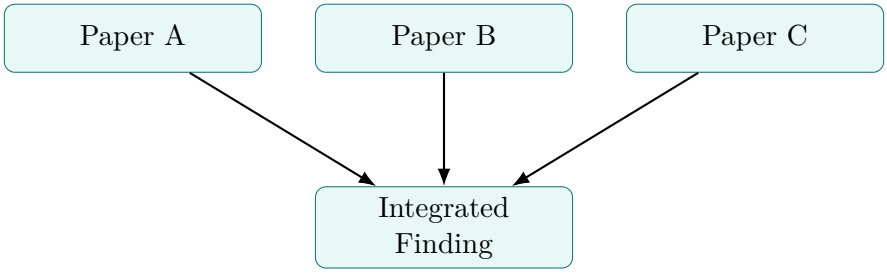
Accuracy: Paper B (96%). Cost: Paper A (low complexity). Real-time: Paper A (good scalability).

11. LITERATURE SYNTHESIS

From Individual Papers to the Field

11.1 What is Synthesis?

Analysis breaks individual studies into components.
Synthesis combines findings across multiple studies to understand the larger research landscape.



11.2 Techniques for Literature Synthesis

Technique	Purpose
Thematic Synthesis	Group studies around common themes
Methodological Comparison	Compare algorithms, datasets and experimental approaches
Chronological Review	Show how research has evolved over time
Conceptual Mapping	Connect major concepts and relationships
Systematic Review	Use a structured process to identify and synthesize literature

11.3 Finding the Research Gap

Possible research gaps may appear when:

- studies produce contradictory findings,
- datasets are outdated,
- real-world validation is missing,
- scalability is ignored,
- cost or computational limitations are ignored,
- ethical aspects are not considered,
- a method works in theory but not in practical deployment.

PRACTICE ZONE

From the three papers discussed earlier, complete:

What has been solved?

ML methods reach high accuracy for intrusion detection on benchmark datasets.

What remains unsolved?

Real-time, scalable detection on modern network traffic is still limited.

Possible research gap:

Low-latency, low-false-positive detection validated on current traffic.

12. LITERATURE MAPPING

Seeing the Big Picture

12.1 What is Literature Mapping?

Literature mapping is the process of systematically or visually organizing research papers to identify:

- themes,
- clusters,
- influential studies,
- relationships between topics,
- research trends,
- research gaps.

12.2 Why Map Literature?

When a researcher has 30, 50 or 100 papers, a simple list becomes difficult to understand.
A map helps answer:

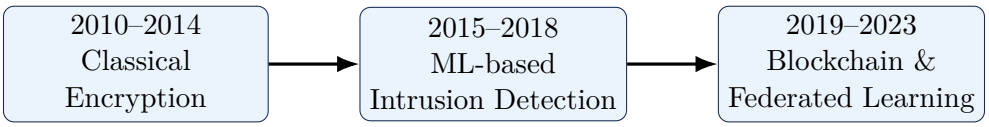
What are the major themes?
Which papers are connected?
How has the topic evolved?
Where is the research gap?

12.3 Types of Literature Mapping

Mapping Type	Description
Thematic	Group papers according to recurring themes
Chronological	Organize research according to time
Methodological	Compare methods such as ML, DL, SVM, CNN, etc.
Citation	Track citation relationships and influential papers
Conceptual	Connect concepts using diagrams or mind maps

13. LITERATURE MAPPING EXAMPLE — IoT Security

Consider the evolution of research on IoT security:



This type of mapping can reveal a shift from classical security methods towards AI-based and decentralized approaches.

13.1 Mapping Tools

Tool	Use
VOSviewer	Keyword, co-occurrence and citation maps
CiteSpace	Citation networks and research trends
Connected Papers	Exploring related research papers
XMind / MindMeister	Manual thematic mapping
Spreadsheets	Simple manual literature organization

CLASSROOM ACTIVITY

Mapping Activity
Take five papers on one topic.
For each paper, identify:

1. Main theme
2. Method
3. Year
4. Important keyword

Then group papers that have something in common.
My theme groups:

Group 1: classical encryption; Group 2: ML-based detection; Group 3: blockchain / federated learning.

14. HANDS-ON LITERATURE MAPPING

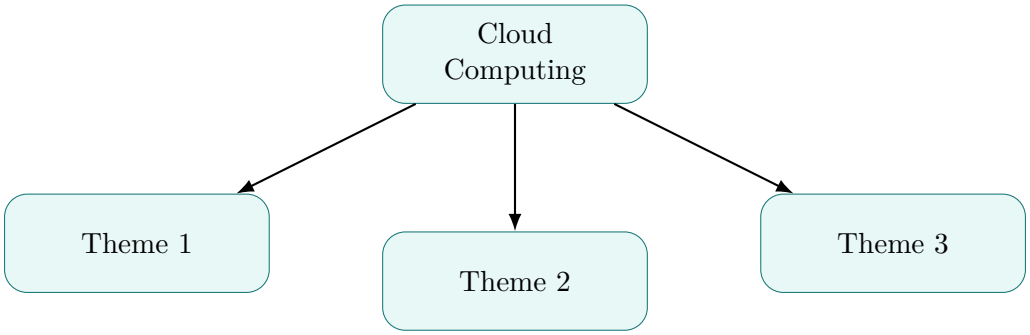
From Papers to Clusters

CLASSROOM ACTIVITY

Topic: Cloud Computing
Assume that you have five papers.
Identify three recurring themes such as:

- Cost efficiency
- Scalability
- Energy consumption

Create your own thematic map below.



Theme 1: Cost efficiency.
Theme 2: Scalability.
Theme 3: Energy consumption.

THINK LIKE A RESEARCHER

What research gap could become visible after mapping these papers?

Few studies jointly optimize cost, scalability, and energy use in cloud systems.

14.1 Best Practices for Literature Mapping

- Start mapping early.
- Combine manual notes with digital tools.
- Update the map periodically.
- Focus on influential work as well as recent work.
- Map ideas and relationships, not only keywords.

15. REFERENCES AND CITATIONS

Academic Credibility

15.1 Why References Matter

References serve several purposes:

- acknowledge original ideas,
- allow readers to verify claims,
- help avoid plagiarism,
- establish scholarly context,
- strengthen the credibility of research.

15.2 Citation Styles

Style	Typical Approach
IEEE	Numbered citations such as [1], [2]
APA	Author–year such as (Smith, 2020)
ACM	Author–year or numeric depending on venue
Chicago / Harvard	Commonly used outside core CS venues

LEARN & REMEMBER

In computer science, IEEE and ACM styles are particularly common. Always check the specific journal, conference or university guidelines before formatting references.

15.3 Citation in Writing

Examples:
Direct citation:
According to Smith [1], deep learning has applications in NLP.
Indirect citation:
Deep learning has been widely applied to natural language processing [1].
Multiple citations:
Several studies have investigated this problem [1]–[4].

COMMON TRAP

Do not mix IEEE, APA and other citation formats randomly within the same document.

16. REFERENCE MANAGEMENT

Manage Literature Systematically

16.1 Reference Management Tools

Tool	Typical Use
Mendeley	Reference management and PDF annotation
Zotero	Browser-based collection and citation management
EndNote	Academic reference management
BibTeX	Reference management for LaTeX documents
RefWorks / Citavi	Institutional reference management

16.2 Why Use Reference Managers?

Manual reference management can cause:

- missing references,
- incorrect author names,
- inconsistent formatting,
- forgotten citations,
- duplicate entries.

Reference managers help organize papers and generate citations according to a selected style.

PRACTICE ZONE

Reference Formatting Activity

Given:

Smith, J. (2021). "AI for Cybersecurity." Journal of Computer Science, 45(3), 120–135.

Write the reference in IEEE style.

[1] J. Smith, "AI for Cybersecurity," J. Comput. Sci., vol. 45, no. 3, pp. 120-135, 2021.

Now write it in APA style.

Smith, J. (2021). AI for cybersecurity. Journal of Computer Science, 45(3), 120-135.

17. CITATION ETHICS

Cite Fairly and Accurately

17.1 Good Citation Practices

A researcher should:

- cite the original source where possible,
- cite relevant work,
- include classical work when important,
- include recent work when appropriate,
- acknowledge contributions fairly,
- maintain consistent formatting.

17.2 Citation Padding

Citation padding occurs when irrelevant references are added simply to increase the number of citations.

THINK LIKE A RESEARCHER

Consider:
"Our algorithm is efficient [1–12]."
However, only two of the twelve papers actually discuss algorithmic efficiency.
Is this good citation practice?
Explain.

No; it cites irrelevant work and misrepresents the support for the claim.

17.3 Common Reference Mistakes

- Mixing citation styles.
- Missing references in the bibliography.
- Listing references that are never cited.
- Incorrect author/year/page information.
- Over-citing trivial facts.
- Citing a secondary source instead of the original source.

18. PLAGIARISM

Academic Integrity

18.1 What is Plagiarism?

Plagiarism means using another person's words, ideas or work without appropriate acknowledgment. It can involve:

- written text,
- figures,
- diagrams,
- computer code,
- datasets,
- ideas,
- previously published work.

Plagiarism may be intentional or accidental.

18.2 Types of Plagiarism

Type	Description
Direct / Verbatim	Copying text word-for-word
Mosaic	Combining copied phrases with your own wording
Self-Plagiarism	Reusing previous work without proper acknowledgment
Paraphrasing Plagiarism	Changing words while retaining the original structure
Accidental Plagiarism	Incorrect or missing citation

18.3 Plagiarism in Computer Science

Examples include:

- copying code from GitHub without credit,
- using another researcher's figure without citation,
- submitting the same project for two courses,
- reusing a previous publication as a new work without proper disclosure.

COMMON TRAP

Plagiarism is not limited to copied paragraphs. Figures, code, data and ideas may also require acknowledgment.

19. PLAGIARISM DETECTION

Similarity is a Warning Signal

19.1 Common Detection Tools

Examples include:

- Turnitin
- iThenticate
- PlagScan
- Grammarly similarity features
- MOSS for source-code similarity
- JPlag for source-code comparison

LEARN & REMEMBER

A similarity report is a tool for investigation. A similarity percentage should not be interpreted without examining what material has been matched and why.

19.2 Identify the Problem

Classify each situation:

1. A student copies a paragraph without citation.
2. A student uses a figure from a paper and cites the paper.
3. A student copies GitHub code and gives no attribution.
4. A student paraphrases an idea and cites the original source.
5. A student submits an old report as a new assignment without disclosure.

No.	Ethical / Unethical?	Reason
1	Unethical	Plagiarism (no citation).
2	Ethical	Properly cited figure use.
3	Unethical	Code plagiarism (no attribution).
4	Ethical	Correct paraphrase with citation.
5	Unethical	Self-plagiarism (no disclosure).

20. PARAPHRASING

Reword + Restructure + Cite

20.1 What is Paraphrasing?

Paraphrasing means expressing another author's idea in your own words and sentence structure while preserving the original meaning.

Good paraphrasing requires:

1. understanding the source,
2. writing the idea independently,
3. changing the structure,
4. preserving the meaning,
5. citing the source.

20.2 Example

Original:

> "Machine learning models require large labeled datasets for training."

Poor paraphrasing:

> "Machine learning systems need big labeled datasets for training."

The sentence structure remains almost the same.

Better paraphrasing:

> Effective machine-learning approaches often depend on extensive annotated datasets during model development [1].

COMMON TRAP

Simply replacing words with synonyms is not sufficient. The idea must be understood and expressed independently.

20.3 Practice

Original:

> "Cloud computing improves scalability but raises security concerns."

Write your own version.

Although cloud computing improves scalability, it also introduces new security risks [1].

Citation marker: [1]

21. ACADEMIC WRITING ETHICS

Originality and Honesty

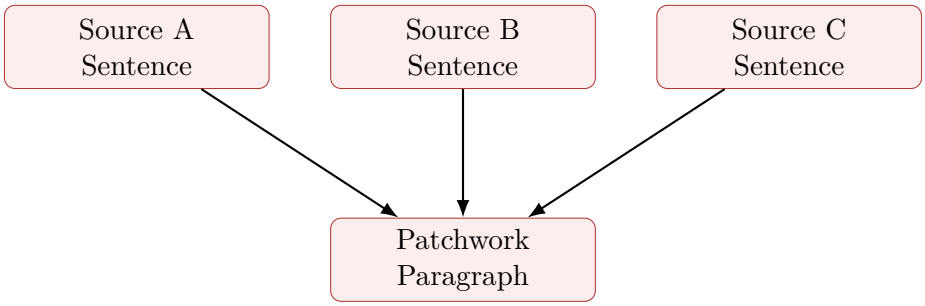
21.1 Ethical Academic Writing

A researcher should:

- cite original sources,
- use quotation marks for direct quotations,
- paraphrase rather than patch-copy,
- acknowledge collaborators,
- acknowledge prior work,
- maintain originality.

21.2 Patchwork Writing

Patchwork writing occurs when a researcher takes sentences or phrases from multiple sources and joins them together with minor changes.



THINK LIKE A RESEARCHER

Why is adding citations to every sentence not automatically enough to make patchwork writing acceptable?

Because the wording and structure are still copied; real writing expresses ideas in your own words.

22. RESEARCH ETHICS

Responsible Conduct of Research

22.1 What is Research Ethics?

Research ethics refers to principles that guide responsible conduct during research.
It promotes:

- honesty,
- transparency,
- fairness,
- responsibility,
- protection of participants,
- reliability of the scientific record.

In computer science, ethics can involve:

- software,
- algorithms,
- datasets,
- artificial intelligence,
- human-computer interaction,
- user studies.

22.2 Core Ethical Principles

Principle	Meaning
Honesty	No fabrication, falsification or misrepresentation
Objectivity	Avoid inappropriate bias in research design and interpretation
Integrity	Follow professional and institutional guidelines
Respect	Protect participants and their information
Accountability	Take responsibility for research work and contributions

23. ETHICAL ISSUES IN CS RESEARCH

People, Data and Algorithms

23.1 Major Ethical Areas

Area	Concern
Human Subjects	Consent, privacy and participant safety
Data Handling	Security, anonymization and lawful use
Authorship	Fair credit for genuine contributions
Publication	No duplicate publication or fake peer review
AI Bias	Avoid unjustified discrimination or harmful bias

23.2 Fabrication and Falsification

Fabrication means creating data or results that never existed.

Falsification means manipulating research materials, data, methods or results so that the research record is misrepresented.

COMMON TRAP

Changing experimental results simply to make an algorithm appear better is a serious ethical violation.

PRACTICE ZONE

A researcher obtains 82% accuracy but reports 94% accuracy in the paper because the higher value appears more impressive.
Identify the ethical violation and explain why it is harmful.

Falsification of results; it misleads readers, wastes others' effort, and corrupts the scientific record.

24. ETHICS IN DATA USAGE

Privacy and Responsible Data

24.1 Responsible Data Practices

When human or sensitive data is involved, researchers should consider:

- informed consent,
- privacy,
- secure storage,
- anonymization,
- pseudonymization where appropriate,
- lawful data collection,
- transparency about data usage.

24.2 Data Privacy Principles

The source material highlights frameworks such as:

- GDPR
- HIPAA
- Indian IT Act
- Digital Personal Data Protection Act (DPDP Act)

LEARN & REMEMBER

A simple principle to remember:

Collect only what is necessary, protect what you collect, and clearly explain how it will be used.

24.3 Privacy Activity

You are developing a mobile application to track student study habits.
Possible data:

- Name
- Age
- Study duration
- Location
- Device information
- Academic performance

PRACTICE ZONE

Which data is actually necessary?

Only study duration and academic performance; drop name, location, and device info.

How would you anonymize the data?

Replace names with random IDs and remove location and device identifiers.

How would you obtain informed consent?

Explain the purpose and data use, then get opt-in agreement before collecting data.

25. AUTHORSHIP ETHICS

Who Should Be an Author?

25.1 Responsible Authorship

Authorship should reflect meaningful contributions to the research.
Examples of significant contributions may include:

- research design,
- substantial experimentation,
- methodology development,
- data analysis,
- substantial interpretation,
- significant manuscript contribution.

25.2 Unethical Authorship

Gift authorship:
Adding a person as an author despite little or no contribution.
Ghost authorship:
A significant contributor is not properly identified as an author.

COMMON TRAP

Being a supervisor, senior person or friend does not automatically justify authorship. Authorship should follow the contribution criteria and the relevant journal/conference policy.

PRACTICE ZONE

Four students work on a project:

A	Research design and data analysis
B	Software implementation
C	Presentation design only
D	Data collection and manuscript preparation

Discuss which contributions could justify authorship.

A, B, and D justify authorship (design/analysis, implementation, data & writing); C (slides only) does not.

26. PUBLICATION ETHICS

Responsible Scholarly Communication

26.1 Common Ethical Violations

Examples include:

- plagiarism,
- self-plagiarism,
- fabrication,
- falsification,
- duplicate publication,
- unjustified authorship,
- fake peer review,
- hiding important conflicts or limitations.

26.2 Ethical Decision Activity

For each situation, identify the main issue.

	Situation	Issue
1	The same manuscript is submitted simultaneously to two conferences.	Duplicate submission.
2	A friend is added as an author without contribution.	Gift authorship.
3	A negative experimental result is hidden.	Selective reporting / falsification.
4	A dataset is used without checking its provenance.	Improper data provenance.
5	A collaborator's major contribution is not acknowledged.	Ghost authorship.

27. RESEARCH ETHICS CASE STUDIES

Learning from Real Situations

27.1 Case Study – Facebook Emotion Study

The source material discusses a large-scale Facebook experiment in which users' news feeds were modified to study emotional effects.

The case raises questions about:

- informed consent,
- participant autonomy,
- privacy,
- ethical approval,
- large-scale experimentation.

THINK LIKE A RESEARCHER

Discuss

If an experiment can affect a person's behaviour or emotional state, what responsibilities should researchers have before conducting it?

Get informed consent and ethics/IRB approval, minimize harm, allow withdrawal, and be transparent.

27.2 Case Study – Data Privacy

The source material discusses cases where large-scale user data created privacy risks.
A key lesson is:

Publicly accessible or apparently anonymized data can still create privacy risks.

PRACTICE ZONE

A researcher downloads a public dataset containing detailed user activity.
Before using it, what should the researcher check?

1. Source and provenance
2. Presence of personal identifiers
3. Terms of use
4. Privacy risks
5. Permission/consent where relevant

Additional consideration:

Check whether individuals can be re-identified even if the data looks anonymized.

28. IRB AND ETHICAL APPROVAL

Protecting Human Participants

28.1 What is IRB?

IRB stands for:

Institutional Review Board

It may also be referred to as an ethics committee.
Its role includes reviewing research involving human participants to ensure that:

- participants are treated appropriately,
- risks are considered,
- consent is handled properly,
- privacy is protected,
- ethical requirements are satisfied.

28.2 When May Approval Be Needed?

Examples include:

- surveys involving people,
- interviews,
- experiments involving participants,
- HCI studies,
- software usability studies,
- studies involving sensitive datasets,
- health, financial or biometric data,
- identifiable personal information.

COMMON TRAP

Do not assume that a project is "only computer science" and therefore automatically exempt from ethical review. Human participants and identifiable or sensitive data can introduce ethical requirements.

29. PRINCIPLES OF ETHICAL APPROVAL

IRB and Participants

29.1 Informed Consent

Participants should understand relevant information about the study, including its purpose, procedures and relevant risks or benefits.

29.2 Confidentiality

Personal information should be protected and not unnecessarily disclosed.

29.3 Minimal Risk

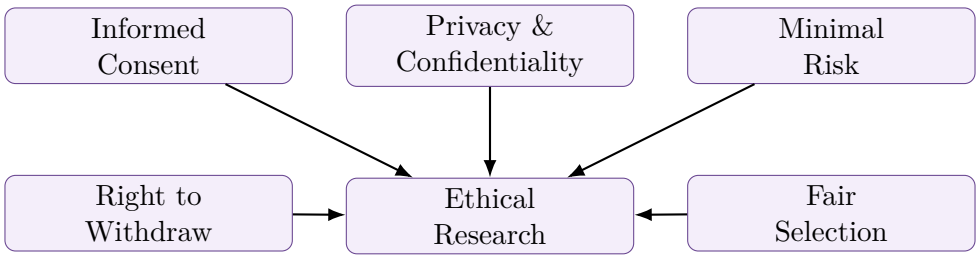
Researchers should minimize avoidable harm, discomfort or risk.

29.4 Right to Withdraw

Where applicable, participants should be able to leave a study without inappropriate consequences.

29.5 Fair Selection

Participant recruitment should be fair and should not create unjustified discrimination or exploitation.



30. INTEGRATED RESEARCH ACTIVITY

From Search to Ethical Research

CLASSROOM ACTIVITY

Choose one research topic from your discipline.
You will now move through the complete Unit III workflow.

Step 1 – Define the Research Problem

Topic:

Reducing false positives in ML-based network intrusion detection.

Research Question:

Can feature selection lower the false-positive rate without reducing recall?

Step 2 – Develop Keywords

Primary keyword	intrusion detection
Synonym	network anomaly detection
Related keyword	false-positive reduction
Technical term	machine-learning classifier

Step 3 – Create Search String

("intrusion detection") AND ("machine learning") AND ("false positive" OR "feature selection")

Step 4 – Select One Important Paper

Title	Feature Selection for ML-Based Network Intrusion Detection
Author(s)	A. Researcher et al.
Year	2021
Database	IEEE Xplore
Method	Random Forest with feature selection
Main Finding	Fewer features cut false positives while keeping recall
Limitation	Evaluated on a single dataset

31. INTEGRATED PAPER ANALYSIS

Analyse, Compare and Identify

Step 5 – Compare Two Papers

	Paper 1	Paper 2
Research Problem	High false positives in IDS	Detection accuracy in IDS
Method	SVM	Deep learning
Dataset	KDD Cup 99	NSL-KDD
Main Result	92% accuracy	96% accuracy
Strength	Simple and fast	High accuracy
Limitation	Outdated dataset	Higher compute cost

Step 6 – Synthesis

Complete:

Across the two papers, the common finding is:

Both apply ML to network intrusion detection on benchmark datasets.

The major difference is:

SVM is simpler; deep learning is more accurate but costlier.

A possible research gap is:

Low-cost methods that still reach high accuracy in real time.

Step 7 – Citation Chaining

One earlier paper I would investigate:

A foundational KDD Cup 99 intrusion-detection study.

One newer paper that cites the selected work:

A recent deep-learning IDS paper citing this work.

32. ETHICAL RESEARCH DESIGN

Apply Research Ethics

Continue with the same research topic.

32.1 Does Your Research Involve People?

Human participants involved?	☐ Yes	☑ No
Identifiable data involved?	☑ Yes	☐ No
Sensitive data involved?	☐ Yes	☑ No
Survey/interview involved?	☐ Yes	☑ No
User study involved?	☐ Yes	☑ No

32.2 Ethical Safeguards

If participants or sensitive data are involved, specify:

Consent procedure:

Use publicly released datasets; no direct human consent needed.

Data protection method:

Store data on encrypted, access-controlled machines.

Anonymization method:

Mask or hash IP addresses before analysis.

Who will have access to the data?

Only the research team.

Potential ethical risk:

Possible re-identification from network metadata.

32.3 Authorship Plan

Who may contribute significantly to the research?

Research Design	Lead student researcher.
Implementation	Student developer.
Data Collection	Student.
Data Analysis	Student.
Writing	Student and co-author.
Supervision / Guidance	Faculty supervisor.

33. UNIT III RESEARCH CHALLENGE

Search → Analyse → Map → Research Ethically

CLASSROOM ACTIVITY

Scenario

You are planning a research project on:

"AI-Based Smart Healthcare Monitoring"

Prepare a mini research-search plan.

A. Literature Search

Three databases I would search:

1. IEEE Xplore
2. PubMed / ScienceDirect
3. ACM Digital Library

Four keywords:

1. smart healthcare
2. remote patient monitoring
3. machine learning
4. wearable sensors

Boolean search string:

("smart healthcare" OR "remote monitoring") AND ("machine learning" OR "deep learning")

B. Literature Analysis

What methods would you compare?

CNN, LSTM, and classical ML for vital-sign prediction.

What datasets would you investigate?

MIMIC-III and wearable-sensor datasets.

What limitations would you look for?

Small samples, privacy limits, and poor generalization.

C. Literature Mapping

Which mapping would be useful?

☐ Thematic ☐ Chronological ☑ Methodological

☐ Citation ☐ Conceptual

Why?

It compares ML methods across studies, which fits a method-focused review.

34. ETHICS APPLICATION

Complete the Researcher Mindset

Continue with the healthcare research scenario.

34.1 Ethical Risks

Identify at least four:

1. Breach of patient privacy.
2. Data misuse or re-identification.
3. Biased models across patient groups.
4. Inadequate informed consent.

34.2 Data Protection

What personal information should be avoided if it is not necessary?

Names, exact addresses, and other direct identifiers.

How will sensitive information be protected?

Encryption, access control, and anonymization.

34.3 Participant Consent

Write a short statement explaining what a participant should know before joining the study.

The study's purpose, what data is collected, how it is used and protected, and their right to withdraw.

34.4 Research Integrity

Complete the statements:

1. I will not fabricate data or results.

1. I will not falsify methods or findings.

1. I will acknowledge all sources and contributors.

1. I will protect participants' data and privacy.

1. I will report limitations honestly and in full.

LEARN & REMEMBER

A strong researcher searches systematically, analyses critically, maps knowledge, cites honestly, protects data and conducts research responsibly. **A strong researcher does not only find information. A strong researcher searches systematically, analyses critically, maps knowledge, cites honestly, protects data and conducts research responsibly.**